Simulation Round 2: Fixes and Extensions

Where we stand after Round 1

Parts 1–3 worked. The same-vendor comovement appeared when it should (average correlation 0.0081, almost exactly the theoretical value $0.04 \times 0.2 = 0.008$), vanished when it should, and spiked on event days for the right vendor only. The detection machinery is sound.

Round 1 also surfaced three problems and one surprise. The problems: (i) our standard errors are too small – the "null" dataset showed a significant same-vendor coefficient (b_1, $t \approx 2.9$) where the true effect is exactly zero, which is a symptom, not a finding; (ii) single-run results can mislead – b_3 came out at $t \approx 2.06$ in one draw where the truth is zero; (iii) the usage slope came in at $b_2 + b_3 \approx 0.165$ when theory says it should be exactly $\theta = 0.2$, an 18% shortfall we need to explain. The surprise: Part 4 tipped to 100% – and that turned out to be the *correct* answer given the numbers we handed you (the quality gap of 0.02 exceeded the boundary value $\approx 0.10 \times \delta^* \approx 0.018$ at which the crowding charge can no longer hold the smaller vendor in the market). Round 2 fixes the three problems and turns the surprise into a figure.

New terms for this round

Clustered standard errors

Our regression treats each firm pair as an independent observation, but firm 17 appears in hundreds of pairs – if firm 17 is unusual, all its pairs are unusual together. Clustering corrects the standard errors for this. Practically: use two-way clustering by (first firm, second firm), or the bootstrap below.

Firm-level bootstrap

Re-sample *firms* (not pairs) with replacement 1,000 times; rebuild the pairs and rerun the regression each time; the standard deviation of a coefficient across re-samples is its standard error. This respects the pair-overlap problem by construction.

Monte Carlo replication

Rerun the entire simulation S times with seeds $1, 2, \ldots, S$ and look at the *distribution* of each estimate. One run can be lucky; two hundred runs cannot.

Oracle residuals

Residuals computed using the *true* β_i we generated, instead of the estimated one. Only possible in a simulation – and useful precisely for that reason: any gap between oracle and estimated results is caused by the estimation step, not the economics.

Fisher transformation

Correlations are squashed into $[-1, 1]$, which slightly distorts regressions on them. The transformation $z = \frac{1}{2} \ln \frac{1+\rho}{1-\rho}$ undoes this. For correlations as small as ours it should barely matter – which is exactly why we check.

Tipping boundary

The quality gap at which the smaller vendor loses its last customer. From the theory it sits where gap $= 0.10 \times \delta^*$; below it both vendors survive, above it the market tips.

Ground rules

Same as Round 1 (seeded, one config file, one script per exhibit), plus one new rule: **no result from a single seed goes in any table**. Every reported number is either a closed-form value or a Monte Carlo summary.

1 Part 5: Fix the inference (one day)

Using the Round 1 design exactly (both datasets, same seeds):

1. Rerun the Part 2 regression three ways: (a) plain OLS standard errors (as in Round 1, for comparison); (b) two-way clustered by both firms in the pair; (c) firm-level bootstrap with 1,000 re-samples.
2. Apply all three to Dataset ZERO as well.

Report: one table – the four coefficients, three sets of standard errors, both datasets.

What you should see: clustered and bootstrap standard errors noticeably larger than plain OLS (roughly 2–4× for b_1); under Dataset ZERO, *no* coefficient significant at 5% once properly clustered. If b_1 remains significant under ZERO with corrected standard errors, stop – there is a bug (most likely in residualization), not a standard-error problem.

2 Part 6: Replace luck with distributions (one day)

1. Wrap the entire Part 1–2 pipeline in a loop over seeds $1, \ldots, 200$, for $\theta \in \{0, 0.05, 0.10, 0.20\}$. For speed you may cut to $N = 300$ firms and 5,000 pairs per run; document the choice.
2. For each run store: b_1, b_2, b_3, their clustered t-stats, and the same-vendor minus cross-vendor average correlation gap.
3. Compute, per θ: the mean and standard deviation of each coefficient across runs; the fraction of runs where b_2 is significant at 5% (this is the test's *power*); and for $\theta = 0$, the fraction where anything is significant (the *false-positive rate*).

Report: Table – coefficient means and Monte Carlo standard deviations by θ; Figure – power of the b_2 test against θ, with the false-positive rate at $\theta = 0$ drawn as a horizontal reference line at 5%.

What you should see: power rising steeply in θ (near 100% at $\theta = 0.2$ with these sizes); false positives at roughly 5% with clustered inference. Round 1's stray significant b_3 should reveal itself as ordinary: about 1 run in 20 will show it at $\theta = 0$.

3 Part 7: Explain the missing 18% (half a day)

Theory says the same-vendor slope on $\delta_i\delta_j$ equals θ exactly. Round 1 measured 0.165 against a true 0.2. Diagnose by a ladder – change one thing at a time, using seed 42 so results are comparable to Round 1:

1. **Oracle residuals.** Rerun Part 2 using the true β_i. If the slope jumps to ≈ 0.2, the shortfall comes from estimating β_i, and we report slopes with a note; if not, continue.
2. **Fisher transformation.** Rerun with z_{ij} in place of ρ_{ij}. Slopes on z should be nearly identical for small correlations; a material change indicates the squashing mattered.
3. **Sample length.** Rerun at $T = 4{,}000$ days. Sample correlations are noisy at $T = 1{,}000$; if the slope walks toward 0.2 as T grows, the shortfall is finite-sample and we report the T-dependence.
4. **Specification.** Drop b_1 and b_3 and regress same-vendor pairs only: $\rho_{ij} = a + \theta\,\delta_i\delta_j$. This removes any collinearity between the level and slope terms.

Report: one table – the estimated slope at each rung of the ladder, with the true value 0.2 as the top row.

What you should see: one rung will account for most of the gap. Whichever it is, that becomes a permanent note in the full protocol: the same distortion will operate on real data, where we cannot compare against a known truth – the simulation is how we learn the correction factor.

4 Part 8: The tipping boundary (one day)

Round 1's "failure" was the model working: at a quality gap of 0.02 the theory itself says the market tips. Now map the whole picture.

1. **Interior rerun.** Repeat Part 4 exactly, but with vendor benefit rates 0.20 and 0.19 (gap 0.01, inside the boundary). Everything else unchanged.
2. **The sweep.** Run the Part 4 loop for quality gaps $0.000, 0.0025, 0.005, \ldots, 0.05$ (21 runs). Record vendor A's final share of firms, both vendors' usage, and total systematic variance $0.2 \times (D_A^2 + D_B^2)$ for each.
3. **Overlay the theory.** On the share-vs-gap plot, draw the predicted line: share difference grows linearly with the gap at rate $1/(0.10 \times \delta^*)$ until it hits 100%, and is flat at 100% after. The kink is the tipping boundary.
4. **The frontier experiment, done right.** Starting from the gap-0.01 interior equilibrium, raise vendor A's benefit rate by 0.005 (small enough to stay interior). Report shares, usage, and systematic variance before and after.

Report: Figure – vendor A's final share against the quality gap, simulated points with the theoretical line and the kink marked; Figure – total systematic variance against the quality gap on the same axis or a twin panel; Table – the interior frontier experiment, before vs. after.

What you should see: (1) simulated points tracking the theoretical line closely, with tipping at a gap ≈ 0.018 – Round 1's parameters sat just past the kink; (2) systematic variance *rising* with the gap even before tipping (mass concentrating on A), then jumping at the boundary; (3) in the frontier experiment, A's share and systematic variance both rise while the market stays split – the project's punchline, now visible because we are inside the boundary. If simulated points deviate from the line away from the kink, check that the crowding charge uses the current iteration's crowd sizes.

Deliverables summary

Part	Deliverable	Purpose
5	Three-way SE table, both datasets	Kill the false positive; certify inference
6	Monte Carlo table + power figure	Replace single-seed luck with distributions
7	Diagnostic-ladder table	Account for the 18% slope shortfall
8	Share-vs-gap figure with theory overlay	The tipping boundary, made visible
8	Systematic-variance figure	Concentration $\Rightarrow$ risk, pre- and post-kink
8	Frontier-experiment table	The punchline, in the interior regime

Suggested pace: four working days plus the memo. Order matters: Part 5 before Part 6 (the Monte Carlo must use corrected inference); Part 7 any time; Part 8 independent. As before, close with a two-page memo in words. One scope note: stay inside this handout – extensions to the economic environment (pricing rules, capacity limits, alternative cost functions) are out of scope for this round and, if suggested, go in the memo's final paragraph as ideas, not in the code.